

I Was Looking for a Job and Then I Found a Job. Heaven Knows I'm $V^E - V^U$ Better off Now

A Discrete Choice Model of Job Search with Reference Dependence and Initial Savings

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1 Introduction

Unemployment exit rates tend to follow a characteristic pattern over the unemployment spell. The hazard rate is relatively high shortly after entering unemployment, declines as the unemployment spell progresses, and increases sharply around the exhaustion or reduction of unemployment benefits. After the spike in hazard rates at the benefit change or exhaustion it falls again. This pattern is difficult to reconcile with the standard job-search model. The standard model predicts a steadily increasing search effort as the benefits decrease and the net value of finding a job rises.

A common way to find the decreasing hazard rates pattern in a standard model is to introduce unobserved heterogeneity. Workers differ in their ability or willingness to find employment. So the most able people quickly find a job, whereas less able people stay longer in unemployment. This changes the composition of the unemployment pool and generates declining aggregate hazard rates. However, this explanation does not account for the sharp changes in search behavior observed around reductions in unemployment benefits.

DellaVigna et al. (2017) propose an alternative explanation based on reference-dependent preferences. Individuals evaluate consumption relative to a reference point determined by recent income. They experience additional utility and disutility if consumption ends at level above or below the reference point. Newly unemployed workers therefore search intensively when their income falls, but relaxes their search effort as the reference point adjusts to a lower income. When workers anticipate a future reduction in unemployment benefits they anticipate the extra loss and that increases the value of finding employment and search effort rises again. The model can therefore generate declining unemployment hazards together with spikes around benefit reductions without relying on unobserved heterogeneity.

A central assumption behind this mechanism is how changes in income translate into changes in consumption. If unemployed workers consume their current benefit income hand-to-mouth, a fall in unemployment benefits translates directly into lower consumption. In reality, however, workers may enter unemployment with savings and use these assets to smooth consumption over the unemployment spell. Savings can therefore work as a form of private unemployment insurance. A worker with sufficient liquid assets may partly

compensate for a reduction in benefits by drawing down savings, such that the fall in income does not translate one-for-one into a fall in consumption. This may reduce the effect of benefit changes on the value of employment relative to unemployment and consequently weaken the response in search effort.

In this paper we extend the job-search framework of DellaVigna et al. (2017) by allowing unemployed workers to enter unemployment with heterogeneous levels of initial assets. Individuals choose both consumption and search effort dynamically and can use savings to smooth consumption throughout the unemployment spell. We compare the model with a standard one-type job-search model, models with heterogeneous search costs, and a reference-dependent specification. This allows us to study whether heterogeneity in initial wealth provides an additional explanation for unemployment-duration patterns and whether the introduction of savings changes the predictions of reference-dependent job search.

Like DellaVigna et al. (2017), we use the 2005 Hungarian unemployment insurance reform to evaluate the models. Prior to the reform, benefits were constant for the first 270 days where they dropped to a lower level the next 90 days. After one year benefits exhausted and social assistance took over. The reform front-loaded the benefits to a two-step system where an initial high rate lasted for 90 days and a lower rate lasted until 270 days. After 270 days the new system followed the old one. Importantly the overall generosity of the reform remained the same before and after the reform. Thus, the reform changed the timing of benefit reductions without changing total benefits. This setting can be used to test the reference-dependent model. It would predict three things, 1) That post-reform hazard rates would be higher around the benefit drop after 90 days, 2) that this difference would decrease and disappear when reference level adapts, 3) that the hazard rates before the reform would be higher before the benefits drop after 270 as this drop is largest in the pre-reform period.

We solve the models numerically using backwards induction and estimate their structural parameters using Simulated Method of Moments. Model-implied unemployment hazards are matched to empirical hazard rates before and after the reform. We estimate the specifications using both L-BFGS-B and LS-TRF in order to examine the sensitivity of the structural estimates to the numerical optimization procedure.

Our results suggest that initial assets have important implications for predicted search behavior. Individuals with higher initial savings search less at the beginning of unemployment because they smooth consumption, but their search behavior converges towards that of workers with fewer assets as their savings deplete. Consumption smoothing is economically realistic, but it also smooths the model responses to benefit changes, which is against the sharp responses observed in the empirical hazard rates. Still we find that the reference-dependent model appears to capture the timing and shape of the hazard rate spikes better than the heterogeneous agents models. However, the heterogeneous-search-cost model obtains the lowest SMM objective values. We also find that similar fitted hazard rates can be obtained with very different estimated structural parameters.

The remainder of the paper is structured as follows. Section 2 presents the job-search model and the numerical solution. Section 3 describes the data and the Hungarian unemployment insurance reform. Section 4 presents the SMM estimation procedure. Section 5 presents the model predictions and estimation results. Section 6 discusses the interpretation and limitations of the results.

2 Job Search Model

This section presents the discrete time dynamic model that we use to estimate job search effort and unemployment hazard rates. The model adds a reference-dependent utility term to the standard model of job search. We use the model proposed in DellaVigna et al (2017).

2.1 Model Environment

We model job seekers in Hungary between January 2002 and December 2008. We use the same simplifying assumptions as DellaVigna et al (2017). Jobs last indefinitely once found and wages are fixed, so that wages before job loss is equivalent to wages after a new job is found. In every period job seekers decide their search effort $s_t \in [0, 1]$ where search effort also is the probability of being employed in period $t + 1$. Search effort is chosen as a trade-off between utility of employment and disutility of searching for employment. Search cost is given as $c_s = \frac{ks^{1+\gamma}}{1+\gamma}$ where γ by construction is the inverse of the elasticity of search effort with regards to the net value of employment. k is the vital parameter for determining search costs.

Job seekers derive utility directly from consumption. Part of prospect theory by Kahnemann & Tversky (1979) suggests that utility is derived from consumption relative to a reference point. Here consumption above the point yields extra utility and consumption below the point gives disutility. People are loss averse so disutility below the point is higher than utility above the point.¹ We follow DellaVigna et al (2017) by not modeling diminishing effects of consumption further away from the reference point.

$$u(c_t) = \begin{cases} v(c_t) + \eta[v(c_t) - v(r_t)] & \text{if } c_t \geq r_t \\ v(c_t) + \eta\lambda[v(c_t) - v(r_t)] & \text{if } c_t \leq r_t \end{cases} \quad (1)$$

We assume log utility so $v(c_t) = \ln(c_t)$. In the gain domain reference dependent utility is weighted with η in the loss domain it is denoted $\eta \cdot \lambda$ where $\lambda > 1$ to denote loss aversion.

Individuals receive income y in every period. Income can be either wages, w , or unemployment benefits of different size, b . Income can be used for consumption, c , or savings, A . Consumption is subject to the budget constraint $c_t = A_t + y_t - \frac{A_{t+1}}{1+R}$ and it is not possible to borrow from future income.

Habit formation theory suggests that reference points are backwards looking.² The optimal consumption reference point would be past consumption. However, then individuals would optimize their current consumption with regards to the effect on future utility from reference points. For feasibility the consumption path is exogenously given by past income. The reference point becomes the average income over $N \geq 1$ periods. This has the attribute of convergence towards the new income level. After N periods with the same

¹See side 11 i DellaVigna

²Page 12, some guy named Constantinides

income the reference point equals the current income.

$$r_t = \frac{1}{N} \sum_{k=t-N}^{t-1} y_k$$

We look at different alterations of the job search model. We look at the standard model where $\eta = 0$ in 1. We also look at a hand-to-mouth consumption model where $c_t = y_t$ and $A_t = 0 \forall t$. For the hand-to-mouth model we implement heterogeneous agents by defining different search cost parameters, k .

2.2 Value Functions

The unemployed faces the problem of maximizing both consumption and search effort with regards to future periods given initial asset level A_t and a discount factor $\delta < 1$. It is described by the Bellman Equation:

$$V_t^U(A_t) = \max_{s_t \in [0,1]; A_{t+1}} u(c_t|r_t) - c(s_t) + \delta \left[s_t V_{t+1|t+1}^E(A_{t+1}) + (1 - s_t) V_{t+1}^U(A_{t+1}) \right] \quad (2)$$

subject to:

$$c_t = A_t + y_t - \frac{A_{t+1}}{1 + R}$$

For the employed the value function is given as:

$$V_{t|j}^E(A_t) = \max_{c_t > 0} u(c_t|r_t) + \delta V_{t+1|j}^E(A_{t+1}) \quad (3)$$

The value of employment is determined degenerately from the reference point which is determined by current period, t and the period in which the employed found a job, j . Value of employment is high when the initial reference point is low.

2.3 Solving the model

We solve the model by obtaining the optimal consumption rule and search effort in every period for every asset level through backwards induction. We use the decision rules to simulate optimal consumption and search effort for individuals starting with different types of assets.

To begin backwards induction we find the terminal (steady state) values of both employment and unemployment, which we find through value function iterations. We also derive the optimal search effort as given by parameters and value of employment and unemployment.

Steady State Values

We use that N periods after employment the reference point becomes stationary at $r = w$ with a given A . Thus, we are dealing with a cake eating problem for a deterministic growing cake in the infinite horizon. The value function can be written as a Bellman operator without time subscripts, with constant reference and no stochasticity.

$$\Gamma(V^E)(A) = \max_{c \leq A+w} [u(c|r) + \delta V^E(A + w - c)]$$

We solve this by iterating over value functions until we reach a steady-state value.³ As $\delta \in [0, 1[$ and the contraction mapping theorem states that if the Bellman operator is on a Banach space then there is a unique steady-state value. According to Blackwell's theorem the unique steady-state solution is also the one that maximizes value.⁴

The steady state value of employment is not enough to find the full value of employment. The unemployed finds a job in period j and the utility derived from employment is dependent on t as timing determines benefits and thus reference dependent utility in the first N periods. We account for this by backwards inducing from the steady-state value going from period $j + N$ back to j , finding the optimal value in every period. This leaves us with one value of finding a job in every period and for every asset level.

$$\begin{aligned} V_t^E(A) &= \sum_{j=t}^{j+N-1} \beta^{t-j} u(c_{t,j}|r_{t,j}) + \delta^N \cdot \sum_{i=j+N}^{\infty} \delta^{i-j-N} u(w + \bar{A}|w) \\ &= \sum_{j=t}^{j+N-1} \beta^{j-t} u(c_{t,j}|r_{t,j}) + \delta^N \cdot \frac{u(w + \bar{A}|w)}{1 - \delta} \end{aligned} \quad (4)$$

Where $\bar{A}$ is a constant slice of assets that you consume every period and the reference point has converged to the wage, $r = w$.

The steady-state value of unemployment in the terminal period is then: $V^U(A) = \frac{u(welfare + \bar{A}|r)}{1 - \delta}$ where the reference point is the social benefits. We find the steady state value of unemployment through value function iterations again, but change the income level and reference point.

Optimal Search Cost

The unemployed have to optimize both consumption and search effort. To solve that the optimal search effort in period t is constructed to be determined directly by parameters and value of employment and unemployment in period $t + 1$. The first order condition of the value of unemployment, (2), yields:

$$\begin{aligned} \frac{\partial V_t^U(A_t)}{\partial s_t} &= 0 \\ \Leftrightarrow \\ c'(s_t^*(A_{t+1})) &= \delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})] \end{aligned}$$

This shows the marginal change in search cost through changes in search effort for given asset level. To find the optimal search effort we take the inverse of the derivative of the search cost function.⁵

$$\begin{aligned} s_t^* &= \mathcal{C}(\delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})]) \\ \text{where } \mathcal{C}(\cdot) &= c'^{-1}(\cdot) \end{aligned} \quad (5)$$

³The model can be solved analytically only in the highly unlikely case that $\delta(1 + r) = 1$.

⁴Blackwell's Theorem:

$\delta(s) = \arg \max_{d \in D(s)} [u(s, d) + \beta \int V(s') p(ds'|s, d) V(s)] = \max_{d \in D(s)} [u(s, d) + \beta \int V(s') p(ds'|s, d) V(s)]$

⁵See full derivation in appendix.

Job Search Effort, Consumption and Value of Unemployment

We solve the model numerically by backwards induction. We plug the steady state values of employment and unemployment into (5) to find the optimal search effort in $T - 1$. We assume that in period T it is no longer possible to find a job. Then we use these optimal search efforts for period $T - 1$ to calculate the optimal consumption/savings level in the same period. Which we again use to calculate the optimal search effort in period $T - 2$ and so forth. This gives us the search effort and consumption decision rules for all asset levels in every period. Finally, we use these decision rules to find the optimal consumption and search effort path for individuals entering unemployment with all asset levels. We also solve a hands-to-mouth model with agents that has varying search cost parameter, k .

We allow for different asset levels on a discrete grid that are all connected to a utility level. We allow for a finer grid for consumption and therefore we interpolate the values of future asset levels that lie outside the grid.

3 Data ⁶

We use the same data as DellaVigna et al (2017). They have administrative data on social security contributions for 4 million individuals in Hungary in the period between January 2002 and December 2008. Data includes the start and end date of UI benefit spells, so that length of unemployment spells can be calculated. The data is restricted to individuals who are eligible for UI benefits through the full benefit spell (360 days), to individuals between 25 and 49 years of age, and to individuals who earn above the 70th percentile among UI claimants.

DellaVigna et al (2017) uses a Hungarian UI benefit reform from 2005 that front loads the benefit path. Before the reform a flat UI rate equivalent to US\$222 was given for the first 270 days. For the next 90 days a rate of US\$114 is given until the expiration of the UI benefit spell. After the reform a rate of US\$342 was given during the first 90 days. For the next 180 days the rate was reduced to US\$171 and during the last 90 days of the UI benefit spell the rate was once again US\$114. Importantly the overall benefit level for the whole 360 days is roughly similar. After the UI benefit spell a household income dependent social assistance benefit is provided.

We have the empirical hazard rates from unemployment every period both pre- and post the UI benefit reform and their standard deviation. We use hazard rates for every 15 days in a 540 days period, both pre- and post the reform. This gives us 35*2 empirical hazard rates as moments as we follow DellaVigna et al (2017) and disregard the first period, because it is on-the-job search.

4 Estimation

We estimate the model parameters using Simulated Method of Moments. Let $m(\xi)$ be the model predicted hazard rates for each period, and $\hat{m}$ denote the vector of empirical moments.

We find the parameters that make our model predicted moments fit the true population moments, m_0 as closely as possible. Thus we can write our moment condition as the distance between the true and

⁶As we use the data from DellaVigna et al (2017) this section is close to section "III.B. Data" in the mentioned paper.

Benefits path, before and after reform

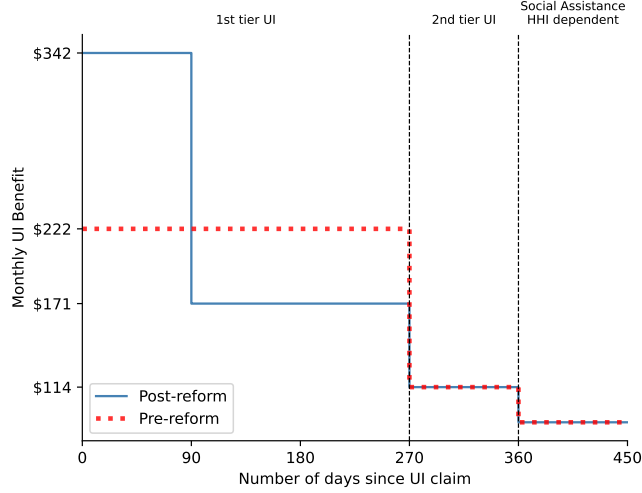


Figure 1: Comparison of predicted hazard rates in the standard and reference-dependent models.

predicted hazard rates, $m(\xi) - m_0$. We use the observed empirical moments, and our estimator is a minimum-distance estimator, that chooses the $\hat{\xi}$ which minimizes the weighted distance between empirical and predicted moments

$$\hat{\xi} = \underset{\xi}{\operatorname{argmin}} (m(\xi) - \hat{m})' W (m(\xi) - \hat{m})$$

W is positive-definite weighting matrix. We follow DellaVigna et. al. in using a diagonal matrix with the inverse of the variance for each moment in its diagonal. Hence, moments with a lower estimated variance, that is moments that are estimated more precisely receive more weight in the estimation.

$$W = \operatorname{diag} \left(\frac{1}{\widehat{\operatorname{Var}}(\hat{m}_1)}, \dots, \frac{1}{\widehat{\operatorname{Var}}(\hat{m}_{70})} \right)$$

We obtain the theoretical moments $m(\xi)$ by following the steps described in section 2 for different vectors of candidate parameters. In the assets model the optimal search efforts for all asset levels are weighted by their share and aggregated to find one accumulated hazard rate. In the heterogeneous agents model we weight agent types by their population share and aggregate the weighted hazard rates.

Under standard regularity conditions, where a parameter value ξ_0 exists as a unique solution satisfying $m(\xi_0) = m_0$, the solution lies in the interior of our parameter space. The moment condition is continuous in ξ and differentiable around ξ_0 , such that the Jacobian, G , exists, is finite and has full column rank. The empirical moments are assumed to satisfy uniform law of large numbers, such that sample moments converge to their population counterparts, and satisfies the central limit theorem, such that

$$\sqrt{N}(\hat{m} - m_0) \rightarrow_d N(0, \Omega)$$

Ω denotes the asymptotic covariance matrix of the empirical moments.

Let $G(\xi) = \frac{\partial m(\xi)}{\partial \xi'}$, denote the Jacobian of the model-implied moments with respect to the parameters. Under these conditions, it holds that $\sqrt{N}(\hat{\xi} - \xi_0) \rightarrow_d N(0, V)$, with the Variance, V , being the standard sandwich

formula

$$V = (G'WG)^{-1}G'W\hat{\Omega}WG(G'WG)^{-1}$$

Then the asymptotic covariance matrix of the estimator is given by:

$$Avar(\hat{\xi}) = \frac{1}{N}(G'WG)^{-1}G'W\hat{\Omega}WG(G'WG)^{-1}$$

We estimate the model by minimizing the simulated method of moments criterion numerically. To check whether the estimated solution is sensitive to the optimization routine, we use two different optimizer algorithms. One is the L-BFGS-B the other is LS-TRF. L-BFGS-B is a limited-memory quasi-Newton method, which is applied to the SMM criterion, whereas LS-TRF is a trust-region least-squares algorithm. For LS-TRF we use a cholesky decomposition of the weighting matrix, $W = LL'$, to rewrite the SMM criterion as the sum of squared weighted moment residuals, $\|L'(m(\xi) - \hat{m})\|^2$. Both algorithms allow parameter bounds, such that each estimated parameter is constrained between a specified lower and upper bound. We compare the estimated parameters and resulting criterion values across the two different optimization routines as a robustness check.

5 Results

This section examines the qualitative predictions of different model specifications, estimates their structural parameters using SMM estimation and and compare their quantitative fit.

5.1 Model predictions

We begin by considering the empirical hazard and survival patterns that the model should be able to predict. The empirical survival rates clearly show that front loading the benefit path leads to lower survival rates until exhaustion of UI benefits. This is primarily due to a large increase in hazard rates after 90 days for post-reform unemployment spells. This is the date where UI benefits fall from USD342 to USD171. The spike in hazard rates for pre-reform unemployment spells sits at the drop in UI benefits after 270 days. Hazard rates in both periods increase right before UI benefits stop.

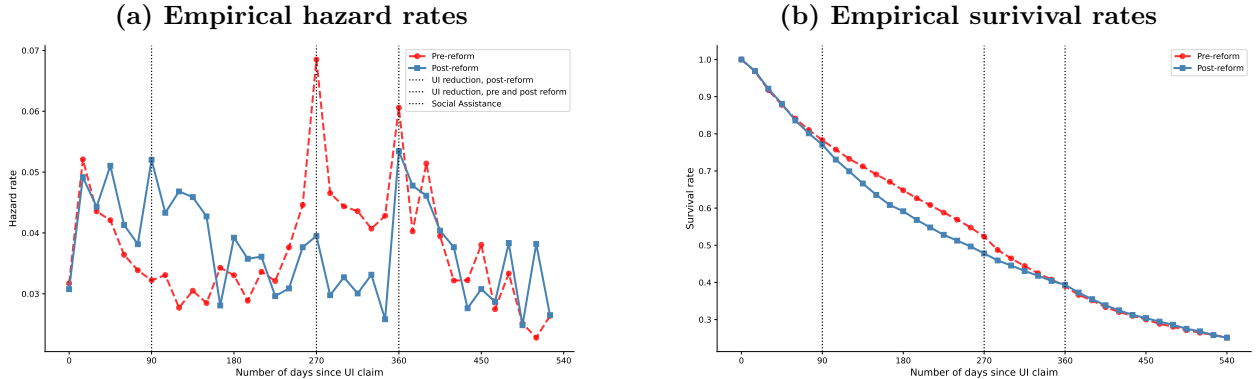


Figure 2: Empirical hazard rates and survival rates

5.1.1 Heterogenous assets

We contribute to the work of DellaVigna et al (2017) by allowing individuals to enter unemployment with different asset levels. Fontaine et al (2024) finds the mean value of liquid assets to be DKK 57.278 roughly equivalent to USD 9000 (exchange rate of 5. september 2026) suggesting small savings to be normal when entering unemployment.

Before estimating the parameters of the reference dependent model with different initial asset levels we output the results for the reference dependent model and standard model with multiple types with the parameters of DellaVigna et al (2017) table 1, see appendix 9.2. We set asset levels to span from USD 0 to USD 17.995. The upper bound is chosen to allow full compensation for the loss in earnings without a job in the pre-reform period, without accounting for interests. Fontaine et al (2024) finds a left-skewed asset distribution in danish job seekers. We assume a similar distribution in Hungary. Asset distribution is shown in appendix 9.3.

Figure 3 presents the model predictions of the standard model with 1 type at different initial asset levels.

- The hazard rates spike at exactly the time of benefit change. Keep in mind that the hazard rate is the search effort from the previous period. This shows that unemployed anticipates the benefit change and thus ramp up their search effort. This is consistent across pre- and post reform predictions.
- Hazard rates are low for unemployed with high asset levels, but converge towards the hazard rates for low asset individuals. This is due to the consumption-smoothing of individuals with assets. Consumption smoothing makes the relative value of employment smaller and decrease job search effort. When assets deteriorate relative value of employment rises and hazard rates converge towards hazard rates of individuals with fewer savings
- The survival rates show that although hazard rates converge individuals with more assets has a higher probability of staying unemployed across all periods, both before and after the reform.
- The consumption paths show that assets allow to keep a higher and smoother consumption, while unemployed without assets follow a hand-to-mouth consumption path
- Assets are used quickly. The convex asset path curves show that the model agent is impatient. For perfectly rational assets would deteriorate in a straight line with jumps at benefit levels.

The standard model correctly predicts changes in hazard rate curvature around periods with benefit change, but it drastically underestimate these changes. Thus, we impose further specifications to find a better fit.

5.1.2 Alternative specifications

Within the hand-to-mouth framework we follow DellaVigna et al (2017) and consider a reference dependent model and a model with unobserved heterogeneity in search costs as alternatives to the standard model.

Reference dependence: We compare the one-type standard model with the one-type reference-dependent model. The reference-dependent term makes the utility function more respondent towards changes in income and thus we expect the reference dependent model to better predict spikes around benefit changes. Figure 4 shows this pattern.

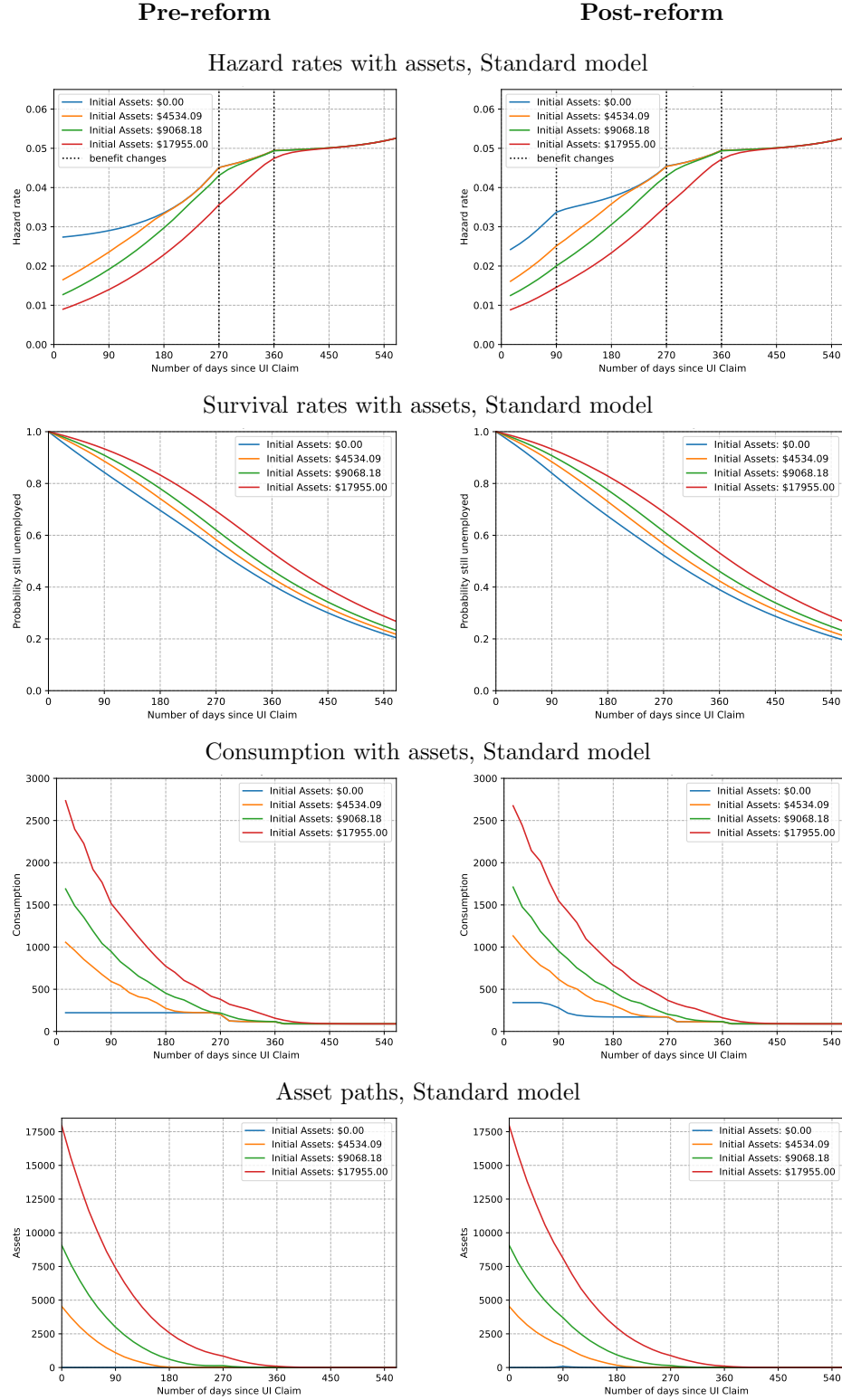
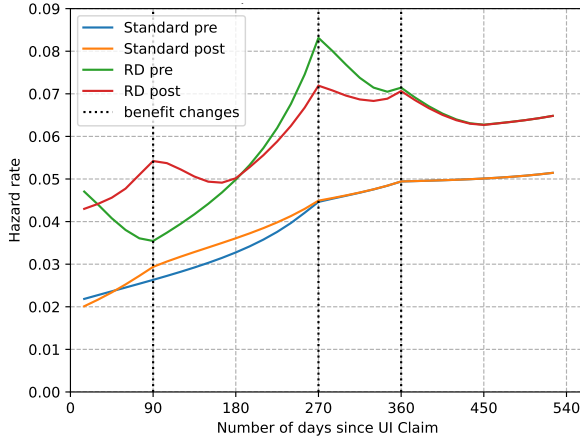


Figure 3: Model predictions for the pre-reform and post-reform period.

The model shows the pre- and post-reform hazard rates, survival rates, consumption and asset paths. The former three is shown at different initial asset levels on the grind of possible assets, while the latter shows how assets develop at different initial asset levels

(a) Reference dependence versus standard



(b) Standard model: 2-type vs 3-type

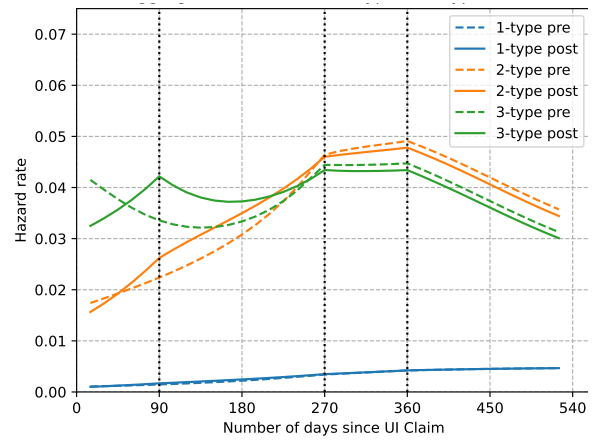
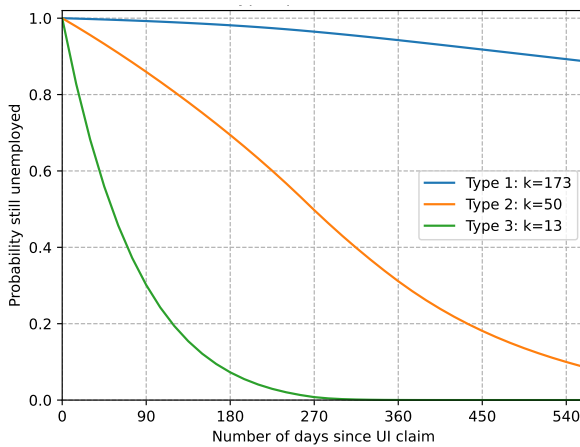


Figure 4: **Hazard rates for the alternative model specifications.** Figure on the left compares predicted pre- and post-reform hazard rates from the one-type standard model and the reference-dependent model, illustrating how reference dependence changes the response to the benefit path over the unemployment spell. The figure on the right compares the predicted pre- and post-reform hazard rates from the two- and three-type standard models, illustrating how additional latent heterogeneity changes the aggregate hazard profile.

Heterogeneous types: We separate unemployed into two and three types with different search costs, low, medium and high. This approach reflects different types that are more or less lazy, skilled or motivated, etc. and thus experience different job search costs. The model predicts higher search effort for those with low and medium search costs, reflecting lower probability of staying unemployed, panel (a), Figure 5, and consequently the population share of unemployed with low and medium search costs are quickly diluted, panel (b), Figure 5.

(a) Type-specific survival rates



(b) Selection across search-cost types

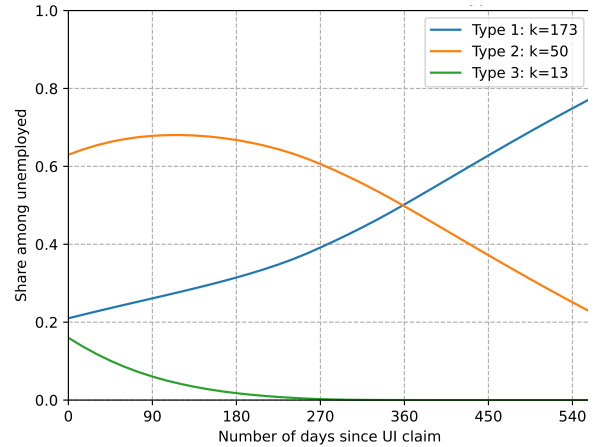


Figure 5: **Type-specific survival and composition of the unemployed.** The left panel shows survival in unemployment by latent search-cost type, while the right panel shows the resulting evolution in the type composition of the unemployed over time

Panel (b) of figure 4 shows how going from a single type to a multiple type model increases the magnitude of the hazard rates mechanically by adding population shares with low search costs.

Interestingly, the 2-type and specially the 3-type model predicts roughly the same spikes as the reference dependent model. This is contrary to our expectations as the agents don't anticipate and aren't affected

as much as the reference dependent agent is. However the net value of employment rises when benefits fall and that should lead to increased search effort among respondent low search cost type. When they enter employment the effects should be smaller as increasing search effort becomes more costly. This is probably why the second spike is large in the reference dependent model compared to the multi type model.

The reference dependence and the 3-type model improves the dynamics of the model, but the mechanisms through which they work are different. The reference dependent model operates through anticipation of falls in benefits. The 3-type model works by allowing different types that follow different trajectories and therefore increase search effort at different times hereby also changing the composition of types left unemployed. Both additions allows for a better model fit to the empirical moments. However, the 3-type model comes with higher costs as we estimate 7 parameters, that is δ, γ , three cost parameters, k_1, k_2, k_3 and two share parameters q_1, q_2 , while with the reference dependent model we estimate 5 parameters $\delta, \gamma, \eta, \lambda, k$.

5.2 SMM estimation

We estimate the heterogeneous asset specification of the model, by matching the model-implied hazard rates to the empirical pre- and post-reform moments using SMM. Table 1 reports the parameter estimates, SMM objective function values (SSE) and computation times for the four model specifications discussed above. These four specifications are estimated twice, once using the L-BFGS-B method and second using the LS-TRF method. Figure 6 prints the model fits of the said models.

The estimated standard models react qualitatively similar to the predicted standard models. The single type model fails to capture the fluctuations around benefit losses and instead plots an increasing hazard rate with tiny curvature kinks around benefit changes. There is no big difference between the estimated pre- and post reform trajectories. The 2-type model captures the increase in hazard rates after 90 days in the post-reform period and the subsequent fall. There is a minimal increase in estimated search effort leading up to the benefit fall after 270 days, but it fails to capture the magnitude of the increase in hazard rates. The 3-type model captures the first benefit change rather well, but as we approach later benefit changes it seems as if low and medium cost types have already left unemployment, and the hazard rates start converging.

The reference dependent model is, to the eye, the model that captures behaviour the best. It acts like a smoothing of the highly varying observed behaviour where it follows the pattern, but doesn't hit the highs, nor the lows of the empirical hazard rates. This generally fits the expectations of the reference dependent model. Reference dependent agents anticipate drops in benefits and act accordingly, but as the new income is being absorbed in the reference period the adjustments become smooth.

Model trajectories almost don't change across the two estimation methods. However, the standard models for 2-types differ. The L-BFGS-B estimation follows roughly the paths by the single type model whereas the LS-TRF estimation follows roughly the paths of the 3-type model.

Table 1 shows the estimated parameter values behind the models portrayed above. The standard model across estimations fit badly, just as it was seen in the graphs. The 3-type standard models both fit similarly well and better than the reference dependent model which is contrary to the intuitive examination of the graph model fits. The reference dependent model however fits far better than the standard model. DellaVigna et al (2017) finds that the reference dependent model performs better than the standard model. This difference is probably due to the introduction of initial assets. The assets also cut the SSE from over 200 to below 100 for the 3-type standard models. We also reduce SSE from around 185 to around 130 for the

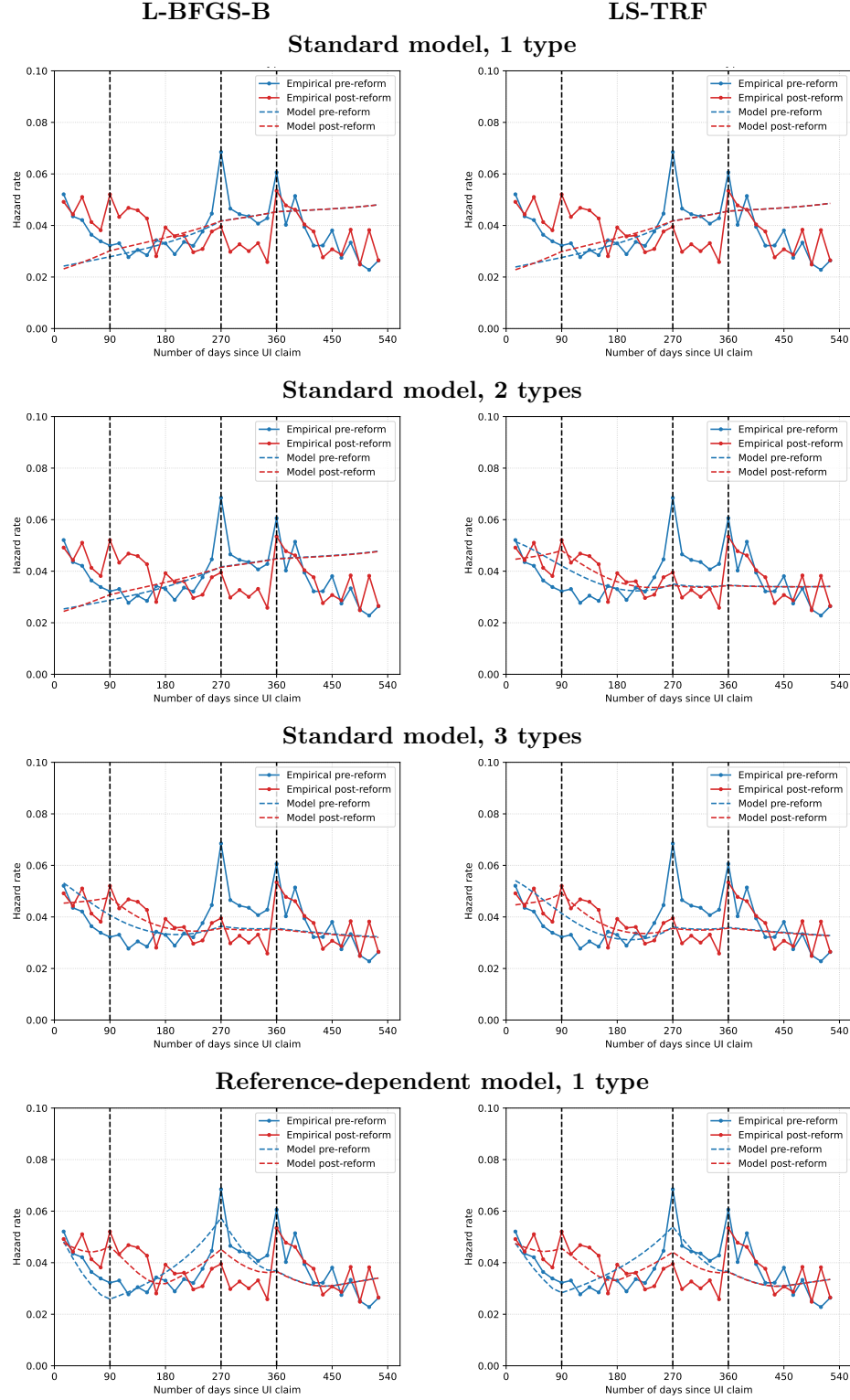


Figure 6: **Model fit before and after SMM estimation.** The left column shows predicted hazard rates evaluated at the initial parameter values, while the right column shows the fitted hazards evaluated at the L-BFGS-B estimates. Rows correspond to the one-, two-, and three-type standard models and the one-type reference-dependent model.

Table 1: SMM estimates across model specifications and optimization algorithms

Parameter	L-BFGS-B				LS-TRF			
	Standard model			Reference dependent	Standard model			Reference dependent
	1 type	2 types	3 types		1 type	2 types	3 types	
δ	0.914	0.942	0.903	0.950	0.917	0.828	0.777	0.950
γ	0.977	0.725	0.795	0.663	0.927	1.363	1.286	0.926
η	—	—	—	1.073	—	—	—	1.384
λ	—	—	—	13.870	—	—	—	14.993
k_1	358.195	300.036	397.627	358.515	315.826	899.938	730.854	894.313
k_2	—	150.523	138.628	—	—	35.918	200.339	—
k_3	—	—	18.259	—	—	—	16.696	—
q_1	—	0.605	0.352	—	—	0.751	0.490	—
q_2	—	—	0.425	—	—	—	0.320	—
q_3	—	0.395	0.224	—	—	0.249	0.190	—
SSE	559.741	517.730	95.522	133.368	578.558	96.627	94.689	126.521
Time (min.)	8.594	32.075	68.317	19.528	1.272	7.821	5.882	2.018

reference dependent models. this reduction comes with a computational price. Comparing run time for the L-BFGS-B estimation shows a difference between the 3-type model and the reference dependent model of 48.8 minutes, in favor of the reference dependent model. However the LS-TRF method reduces run time drastically to a matter of minutes, with the 3-type model running at 5.9 minutes and the reference dependent model at 2 minutes. The Goodness of fit across the 2-type, 3-type and the reference dependent model are all improved as well, although we reach a fairly big SSE for the 2 type model, using the L-BFGS-B method. We suspect that this is due to the minimizer algorithm being stuck in some local minima.

The delta parameter in the standard models fluctuates between 0.78 and 0.94, with the L-BFGS-B method suggesting discount rates around 0.9 to 0.94 for the standard models. Meanwhile, the variation in discount rates across the standard models for the LS-TRF method is between 0.78 and 0.92. However our models maximum discount factor is achieved in both reference dependent models with annual discount factors of 0.95.

This is consistent with the impatience found in DellaVigna et al (2017). They introduce a naive present-biased agent to the model. This both improves the model fit and increase the estimated discount factors to more plausible levels.⁷. A low discount factor makes the model trajectories more respondent to changes in benefit levels. As the standard models don't have the reference term they need a lower delta value to capture spikes. However, this feature indicates that the reference dependent model more realistically captures human behaviour.

The gamma parameter also differs a lot between models and estimation methods. Low gamma values would lead to more responsiveness towards change in incentives, as the elasticity of net value of employment is given as $1/\gamma$. So, low gamma-values should ceteris paribus lead to larger spikes around benefit changes. DellaVigna et al (2017) uses the sanity check of how much increasing the benefit level would change the search effort. This is mostly relevant for extremely high gamma values where small increases in benefit levels could lead to a complete halt in job search. Our gamma values don't differ much from 1 and thus individuals are responsive to changes inside an economic feasible boundary.

⁷We implemented this in our model, but haven't optimized it, as it required optimizing a different part of the model, which we haven't done due to lack of time.

The search cost parameters differ a lot. In the multi-type models there are large differences in k between models and also inside the 2-type and 3-type model. In the LS-TRF estimation it is 44 times more costly for high cost individuals to search for jobs than for low cost individuals. This is in line with findings in DellaVigna et al (2017), but it seems unreasonable that it is 44 times less costly for 19 pct. of the population to search for any job.

We find loss aversion parameters at 14-15. This is 3 times higher than most models in DellaVigna et al (2017) except for a model where they introduce auto regressive reference. High loss parameters makes reactions towards decrease in benefits stronger and it could be a good reason for the reference dependent model to hit the spikes so much cleaner. However, as we have linear loss aversion the parameters seem high.

An additional important point is that the parameter values aren't robust towards changing optimization method. The discount factors in the multi-type standard models fall by more than 0.1 between the L-BFGS-B estimation and the LS-TRF estimation. Equivalently γ rises by around 70-90 pct. for the multi type model and the search cost parameter k is also largely varying. This highlights, alongside the graphs, that you can get to the similar search rates through many different routes.

6 Discussion

We found that a specific model can fit the average level of hazard rate well without necessarily reproducing similar shape and timing of the empirical search responses to benefit changes. The objective function $\hat{\xi} = \operatorname{argmin}_{\xi}(m(\xi) - \hat{m})'W(m(\xi) - \hat{m})$ puts more weight on moments that are estimated more precisely in accordance with our weighting matrix. Thus the criterion for our SMM estimation essentially is a measure of the overall model fit, but might fail to capture distinctive economic features of the model. Hence, to evaluate the model we must combine the SMM objective value with an inspection of the graphical fit. The graphical fit of the model indicates how well it captures economically relevant patterns. The 3-type model performs significantly better than the reference dependent model evaluated solely on SSE. But it does so without capturing the spike in hazard rates after 270 days of unemployment, whereas this second spike is clearly captured in the reference dependent model. The increasingly worse fit of the 3-type model is probably due to the attrition of the low and medium cost types over the unemployment spell. The prediction of the model will be worse, when it becomes similar to a 1- or 2-type model which happens later in the unemployment spell. The reference dependent model fits better throughout the whole unemployment spell, although it hardly captures the last spike. So, our analysis finds that a worse overall fit can still give a better fit of the relevant economic features of the hazard rate, such as timing or shape.

Second, in the reference-dependent model we estimate loss aversion, λ , much higher than DellaVigna et al. (2017). We expect that the higher loss aversion happens mechanically to compensate for the decrease in loss of consumption when assets is added to the model.

Third, the model simulations in section 5.1, shows that high asset individuals entering unemployment, search less because they are able to smooth consumption. Once their assets has deteriorated they enter the search path of individuals without initial assets. The ability of individuals to smooth consumption makes

them less likely to behave as strongly as the empirical unemployed Hungarians in face of a reduction in benefits.

Another problem for the empirical reliability of our model is the underlying asset distribution. We draw loosely on Fontaine et al (2024) to assume a distribution. It probably doesn't represent the actual assets of unemployed in Hungary. A bigger project could use Hungarian wealth data to find the true distribution of assets. It is also paramount to understand which assets are used for consumption smoothing during unemployment. Fontaine et al (2024) finds that most wealth is tied up in illiquid assets and hard to use for consumption smoothing during unemployment spells.

If the unemployed have lower asset levels or the distribution is even more skewed than what we assume, then our model would produce less consumption smoothing and potentially larger asset spikes.

Another approach for multi-type models is an assumption of some correlation between asset-types and search-cost types.

Finally, our estimated parameters appear sensitive to the choice of estimation routine chosen. Although L-BFGS-B and LS-TRF produces similar criterion function values and to some extent similar fitted hazards, the underlying parameter estimates vary substantially. In particular this is evident for the estimates of discount factor, δ , curvature of search cost, γ , and search costs in the heterogeneous type models. The fact that different parameter values produces similar hazard rates and model fit suggests an issue with identification of the individual structural parameters. In other words, parameter values can offset each other and similar hazard rates can be found with very different underlying behaviour. This is mainly a sign of weak identification. A likely issue is that the objective function is relatively flat along some dimensions of the parameter space, which would make a global minimum difficult to find, or at least difficult to distinguish from other nearly identical model fits. If this is the case, then our parameter estimates are not stable. As a consequence our estimated hazard rates appear relatively robust, but we should be careful on interpreting the individual parameter estimates, since they aren't robust to changing estimation method.

7 Conclusion

This paper extends the discrete job-search model in DellaVigna et. al. (2017) by allowing individuals to enter unemployment with assets. We compare different model specifications featuring one or multiple search cost types and a reference-dependent specification in the heterogeneous assets setting. The models are estimated using the simulated method of moments using pre- and post-reform hazards rates from unemployment in Hungary.

We find that to let individuals enter unemployment with different initial asset levels they smooth consumption according to their asset levels during their unemployment spell. It means that workers face a smaller net gain of employment which reduces their search effort. As assets deplete the hazard rates of workers with high initial assets starts converging towards those with no savings.

The 1-type standard model performs poorly in fitting the model, while the models with heterogeneous types and the reference-dependence model improve the fit. The standard multi-type models obtain the lowest SSM objective values, but the reference-dependent specification reproduces the qualitative movements around

benefit changes more closely. We also find that the structural parameters are sensitive to the estimation routine. This suggests that the model lacks in identification of at least some of the underlying parameters of search behavior.

Lastly, our extension of asset levels don't follow the empirical distribution of assets held by unemployed. This is a blow to the empirical reliability of our prediction. But introducing assets have managed to decrease the SSE value to that of DellaVigna et al (2017) anyway and therefore future project of a larger scope could incorporate assets and data for the true asset distribution.

8 Literature

DellaVigna, Stefano, Attila Lindner & Balász Reizer (2017). Reference-Dependent Job Search: Evidence from Hungary. *The Quarterly Journal of Economics*, 10.1093/qje/qjx015.

Fontaine, François, Janne Nyborg Jensen & Rune Vejlin (2024). Wealth, Portfolios and Nonemployment Duration. *Journal of Money, Credit and Banking*, 10.1111/jmcb.13058.

9 Appendix

9.1 Solving for optimal search effort

The search cost is given by $c(s) = \frac{ks^{1+\gamma}}{1+\gamma}$ and thus the derivative of the search cost is $c'(s) = ks^\gamma$. We find

the inverse: $\mathcal{C}(\cdot) = c'^{-1}(\cdot) = \left(\frac{c'}{k}\right)^{\frac{1}{\gamma}} = s$.

The first order condition of the value of unemployment yields:

$$\begin{aligned} \frac{\partial V_t^U(A_t)}{\partial s_t} &= 0 \\ \Leftrightarrow \\ -\frac{\partial c(s_t)}{\partial s_t} + \delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})] &= 0 \\ \Leftrightarrow \\ c'(s_t^*(A_{t+1})) &= \delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})] \end{aligned}$$

From where we take the inverse function to find the optimal s .

$$\begin{aligned} s_t^* &= \mathcal{C}(\delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})]) \\ s_t^* &= \left(\frac{\delta}{k} [V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})]\right)^{\frac{1}{\gamma}} \end{aligned} \quad (6)$$

From here it is clear how γ is the inverse of the elasticity of search effort wrt. to the net value of employment. It also shows that optimal search effort decreases with search costs and increases as future utility matters more and as the net value of employment rises.

9.2 Parameter values in DellaVigna et al (2017)

Table 2: Parameter values for different models

Parameter	Reference Dependent	Reference dependent resent biased	Standard 3-type
Discount factor, δ	0.89	0.995*	0.93
Present bias, β^*	1	0.58	1
Cost curvature γ	0.81	0.37	0.35
Reference dependence, η	1.0	1.0	—
Loss parameter, λ	4.9	4.54	—
Search cost parameter, k	359.0	108.4	—
k_1	—	—	173.1
k_2	—	—	49.6
k_3	—	—	12.9
k1 share, q_1	—	—	0.33
k2 share, q_2	—	—	0.33
N	6	6	6

* In our simulation we set the upper boundary of $\delta = 0.95$, because convergence in VFI isn't reached with higher values.

9.3 Asset distribution

Figure (A1) - Asset distribution in models with assets

